



(<http://www.pieriandata.com>)

Grids

Grids are general types of plots that allow you to map plot types to rows and columns of a grid, this helps you create similar plots separated by features.

```
In [22]: import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

```
In [27]: iris = sns.load_dataset('iris')
```

```
In [28]: iris.head()
```

```
Out[28]:
```

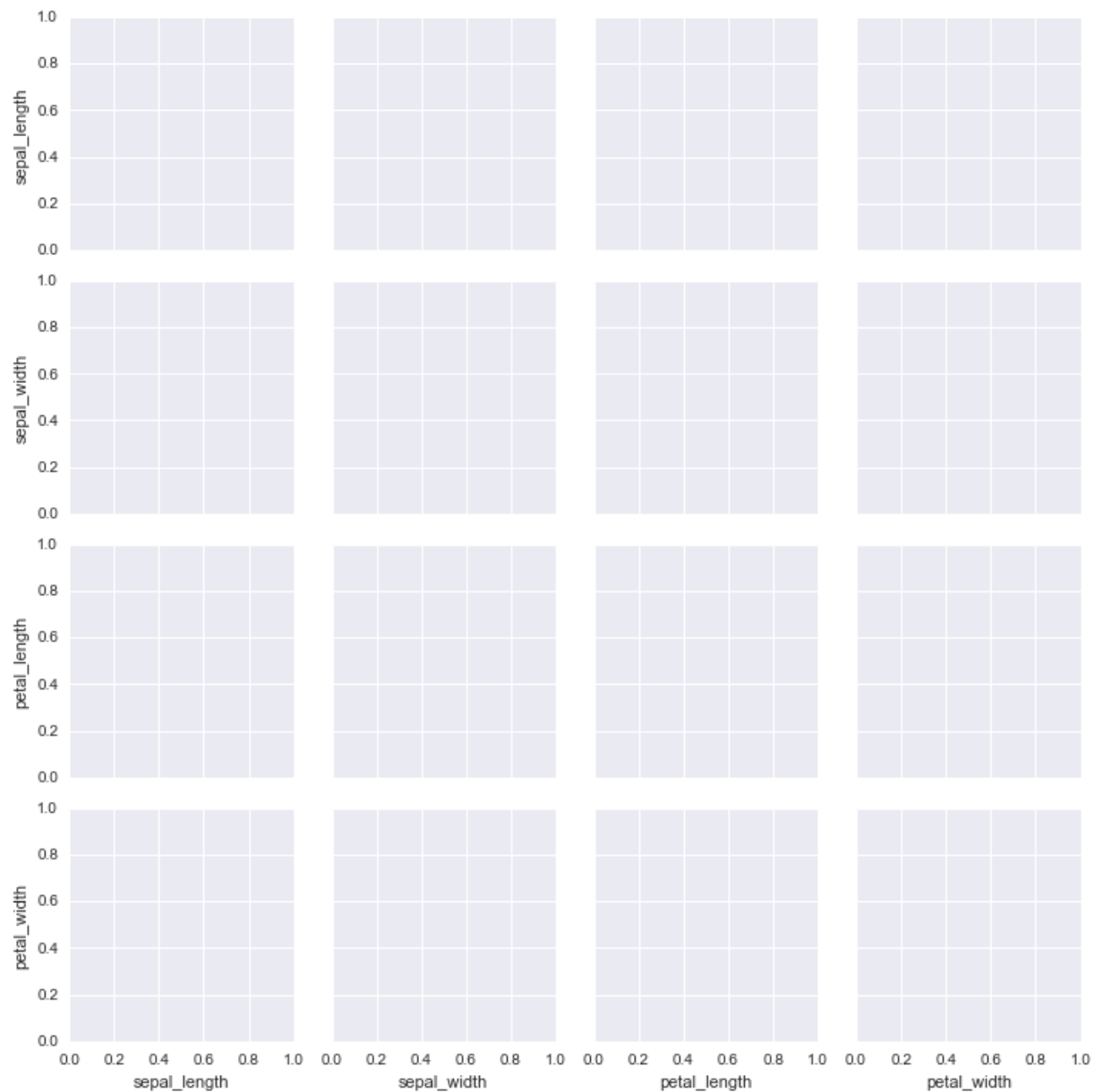
	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

PairGrid

Pairgrid is a subplot grid for plotting pairwise relationships in a dataset.

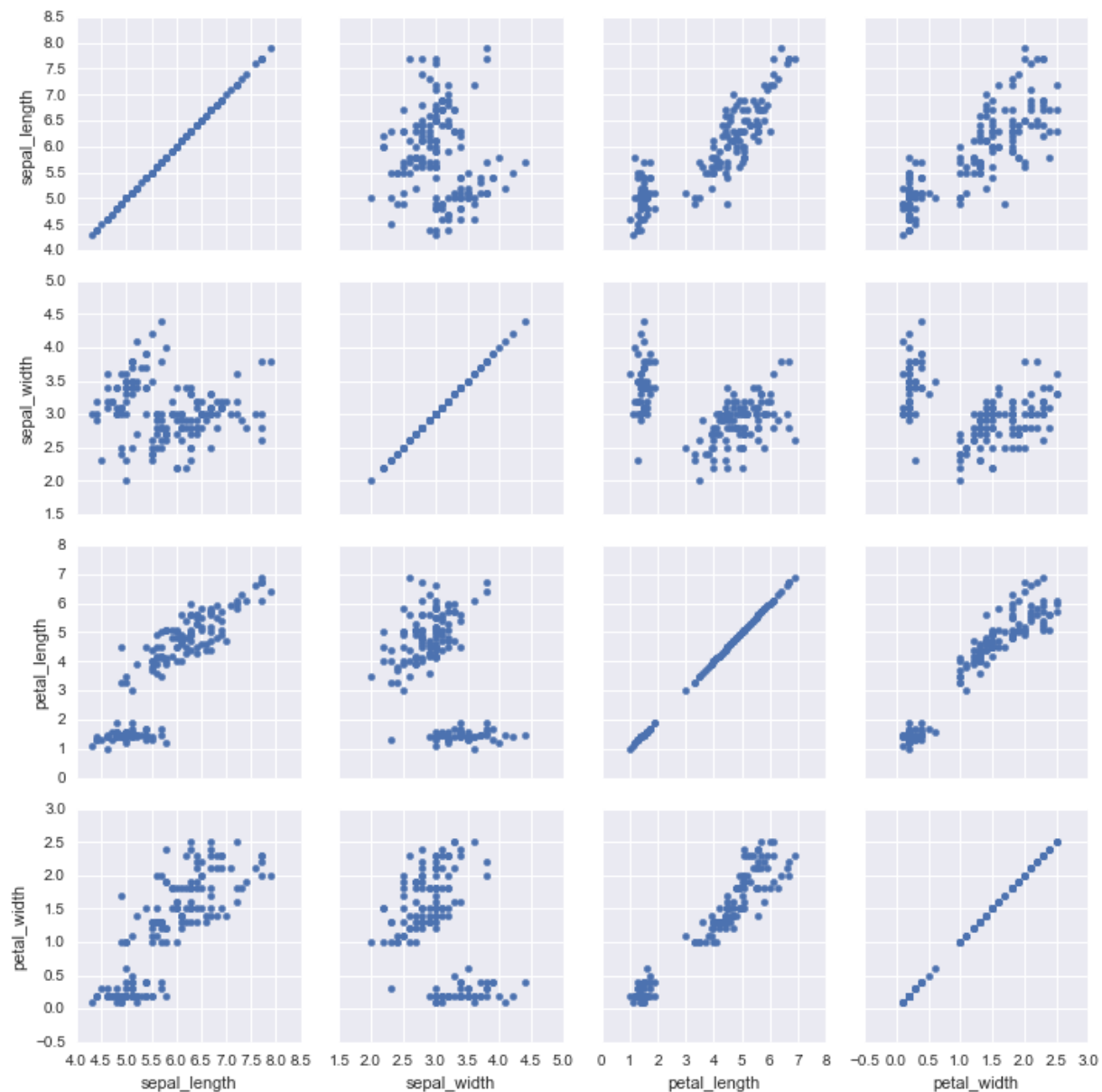
```
In [25]: # Just the Grid  
sns.PairGrid(iris)
```

```
Out[25]: <seaborn.axisgrid.PairGrid at 0x11e39bb38>
```



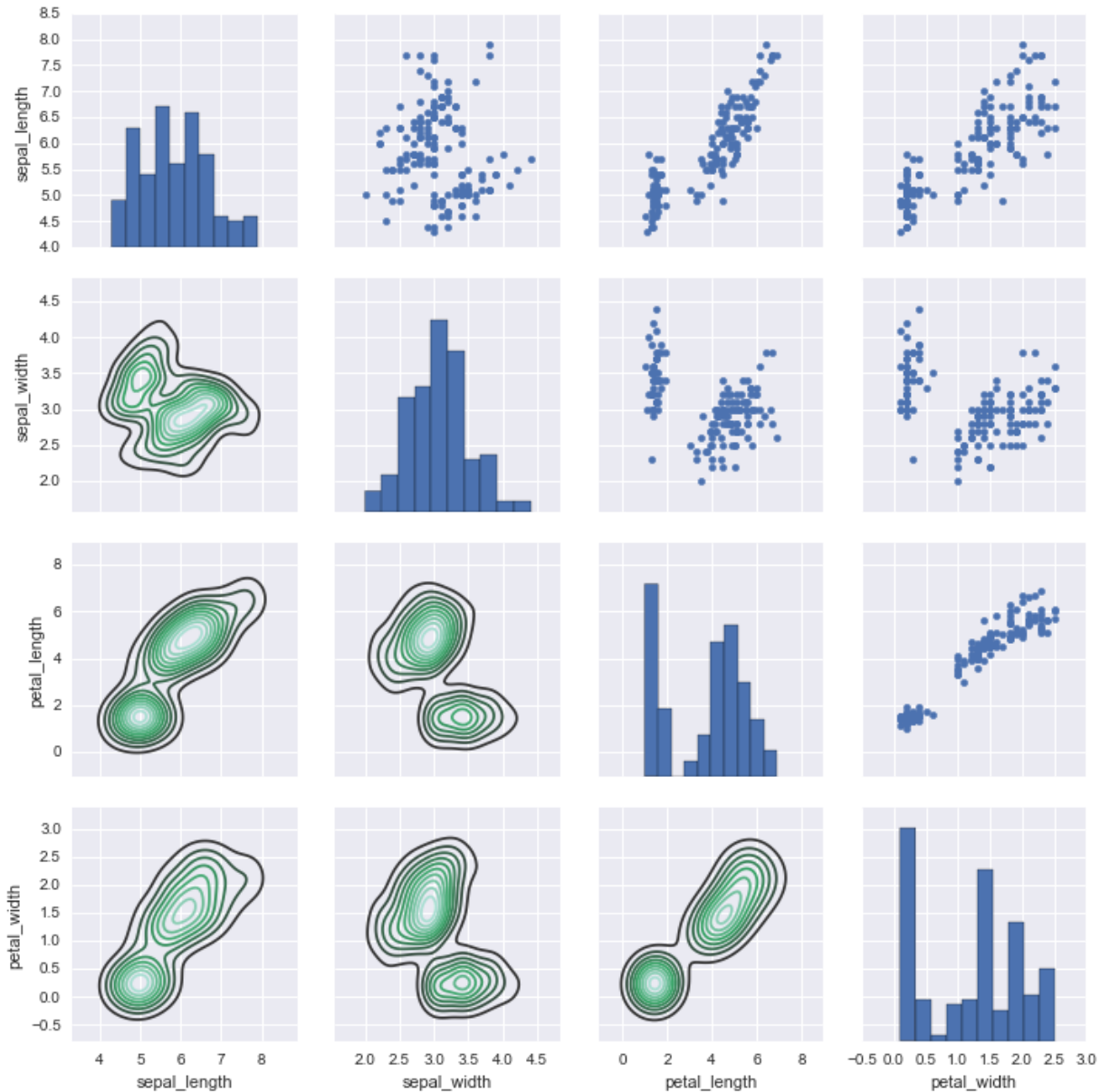
```
In [26]: # Then you map to the grid  
g = sns.PairGrid(iris)  
g.map(plt.scatter)
```

Out[26]: <seaborn.axisgrid.PairGrid at 0x11f431208>



```
In [30]: # Map to upper, lower, and diagonal
g = sns.PairGrid(iris)
g.map_diag(plt.hist)
g.map_upper(plt.scatter)
g.map_lower(sns.kdeplot)
```

Out[30]: <seaborn.axisgrid.PairGrid at 0x121944128>

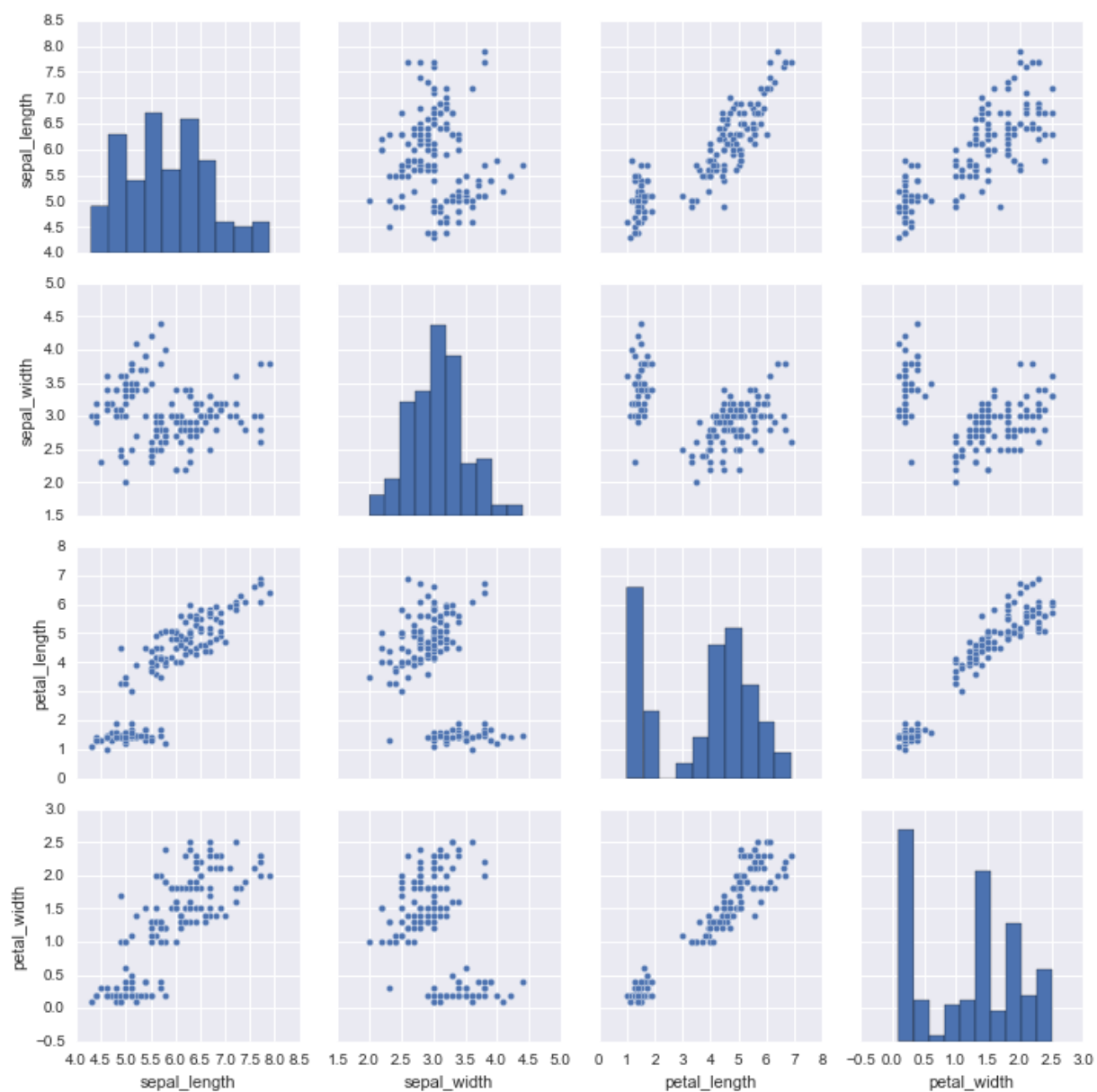


pairplot

pairplot is a simpler version of PairGrid (you'll use quite often)

```
In [31]: sns.pairplot(iris)
```

```
Out[31]: <seaborn.axisgrid.PairGrid at 0x12024b5c0>
```



```
In [33]: sns.pairplot(iris,hue='species',palette='rainbow')
```

```
Out[33]: <seaborn.axisgrid.PairGrid at 0x12633f0f0>
```



Facet Grid

FacetGrid is the general way to create grids of plots based off of a feature:

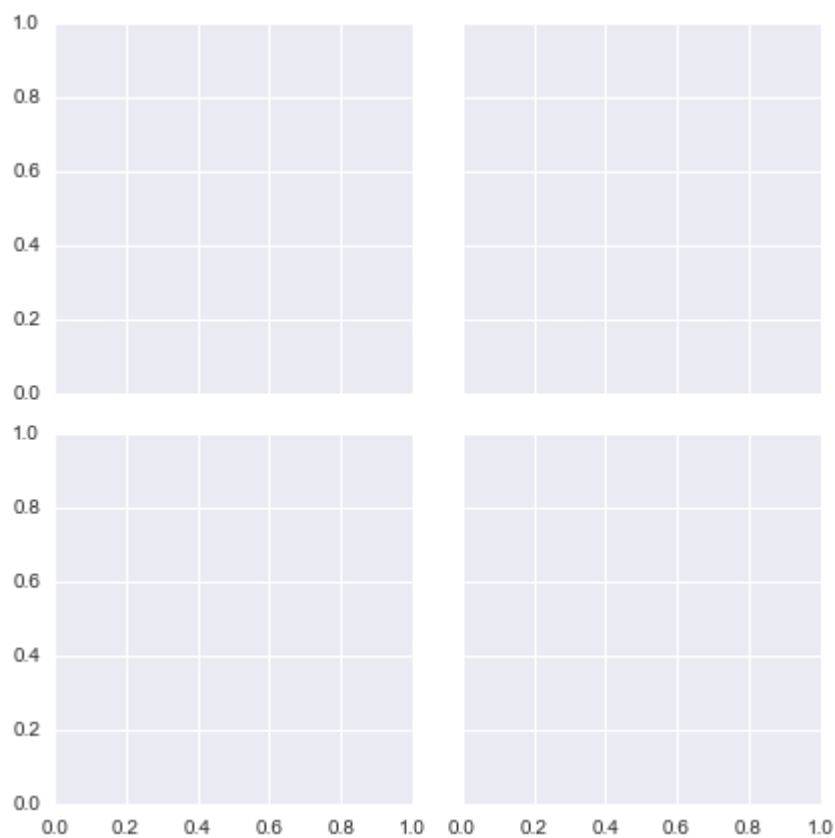
```
In [34]: tips = sns.load_dataset('tips')
```

```
In [35]: tips.head()
```

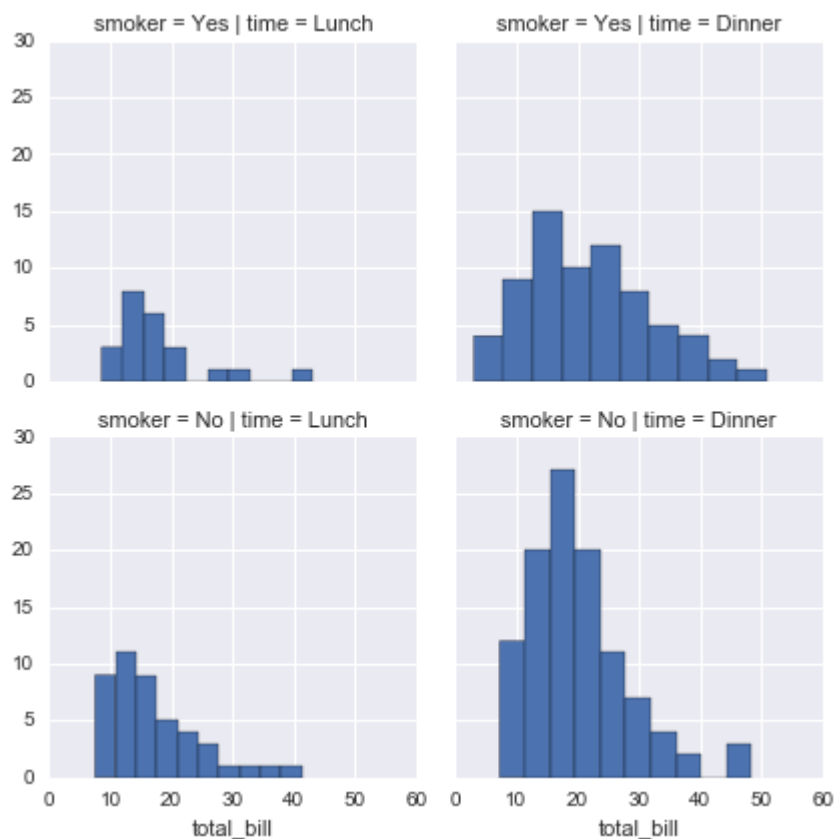
```
Out[35]:
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

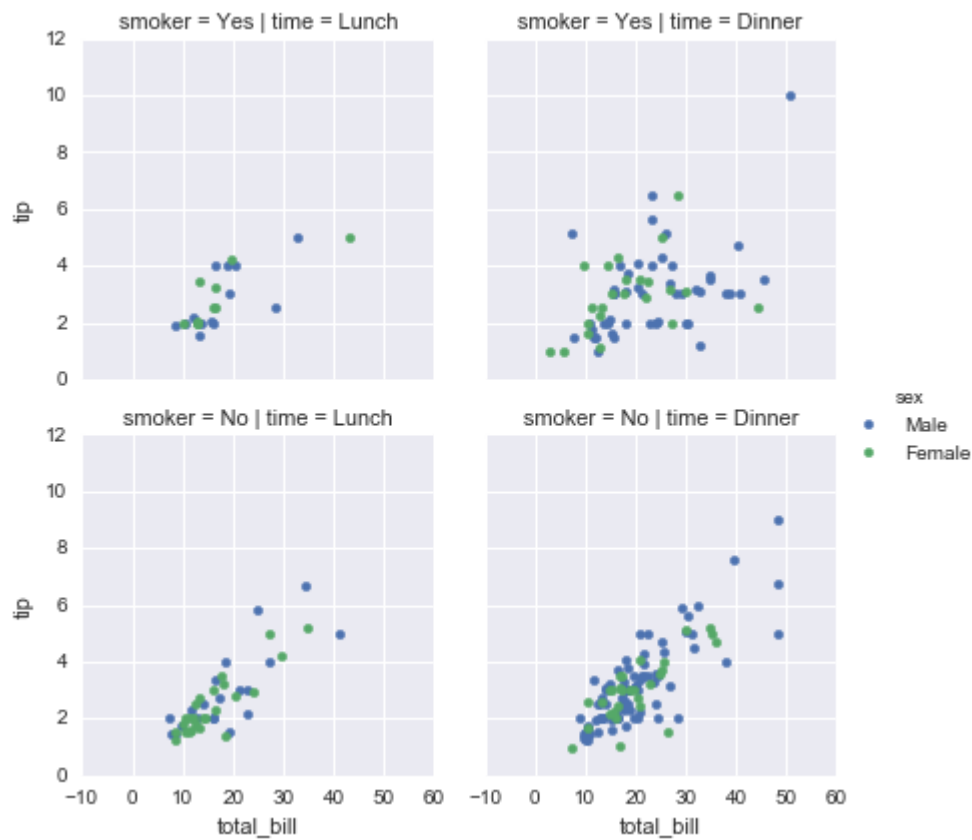
```
In [36]: # Just the Grid  
g = sns.FacetGrid(tips, col="time", row="smoker")
```



```
In [37]: g = sns.FacetGrid(tips, col="time", row="smoker")  
g = g.map(plt.hist, "total_bill")
```



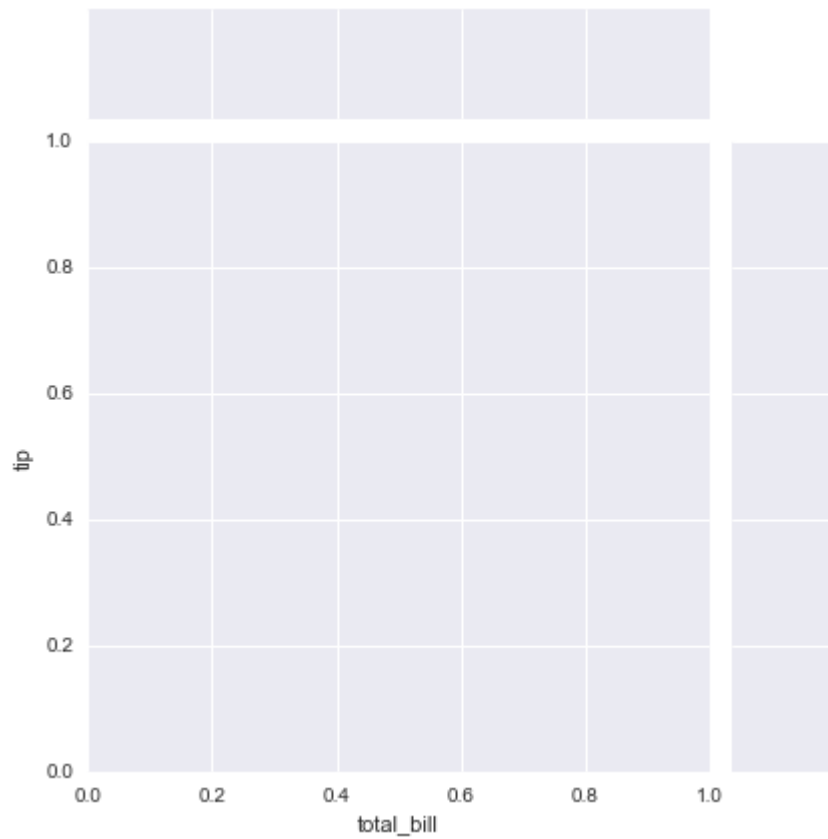

```
In [42]: g = sns.FacetGrid(tips, col="time", row="smoker", hue='sex')
# Notice how the arguments come after plt.scatter call
g = g.map(plt.scatter, "total_bill", "tip").add_legend()
```



JointGrid

JointGrid is the general version for jointplot() type grids, for a quick example:

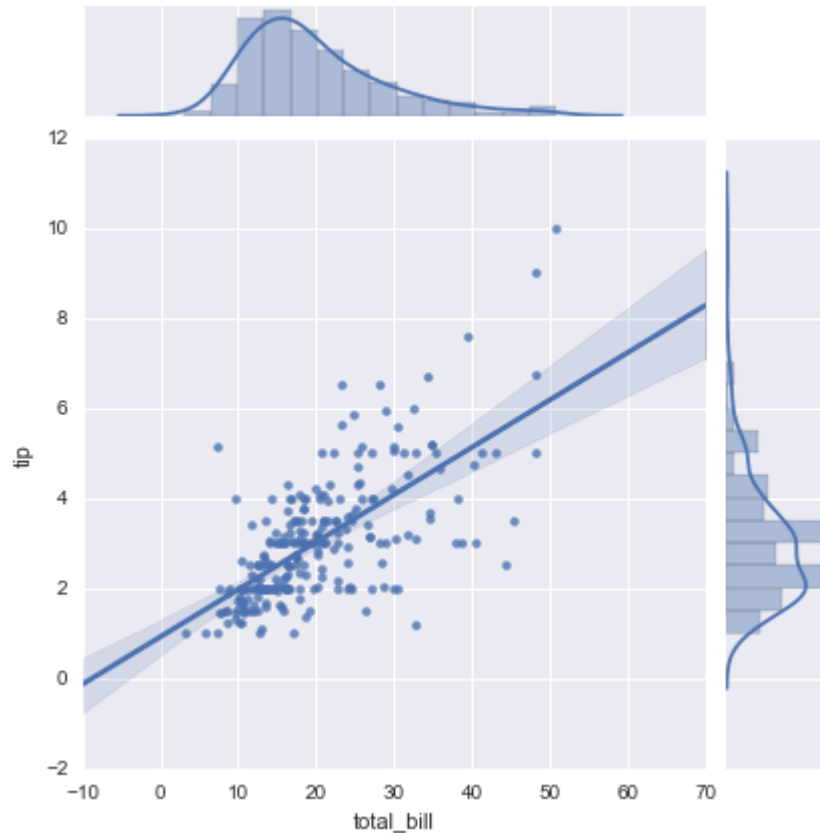
```
In [43]: g = sns.JointGrid(x="total_bill", y="tip", data=tips)
```



```
In [45]: g = sns.JointGrid(x="total_bill", y="tip", data=tips)
g = g.plot(sns.regplot, sns.distplot)
```

/Users/marci/anaconda/lib/python3.5/site-packages/statsmodels/nonparametric/kde
tools.py:20: VisibleDeprecationWarning: using a non-integer number instead of a
n integer will result in an error in the future

```
y = X[:m/2+1] + np.r_[0,X[m/2+1:],0]*1j
```



Reference the documentation as necessary for grid types, but most of the time you'll just use the easier plots discussed earlier.

Great Job!